Assignment 3: From Pthreads to OpenMP

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Task 1: Array Sum

```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/A4$ ./arraysum
for t in 1 2 5 7 9; do ./sumT $t | grep Time; done
for t in 1 2 5 7 9; do ./omp_sum $t | grep Time; done
Sequential array sum, N = 1000000000
Total sum = 500000000500000000 (correct)
Time taken: 0.233802 seconds
Time taken: 0.227460 seconds
Time taken: 0.115576 seconds
Time taken: 0.081312 seconds
Time taken: 0.079896 seconds
Time taken: 0.080210 seconds
Time taken: 0.219171 seconds
Time taken: 0.110991 seconds
Time taken: 0.079512 seconds
Time taken: 0.078042 seconds
Time taken: 0.080231 seconds
```

Task 2: Matrix Multiplication

```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/A4$ ./mainM
for t in 1 2 5 7 9; do ./matrixT $t | grep Time; done
for t in 1 2 5 7 9; do ./omp_matrix $t | grep Time; done
Sequential matrix multiplication, N = 1000
C[0][0] = 1000, C[999][999] = 1000 (expected 1000)
Time taken: 0.567312 seconds
Time taken: 0.480858 seconds
Time taken: 0.259124 seconds
Time taken: 0.132450 seconds
Time taken: 0.119183 seconds
Time taken: 0.107818 seconds
Time taken: 0.483619 seconds
Time taken: 0.246809 seconds
Time taken: 0.122828 seconds
Time taken: 0.107635 seconds
Time taken: 0.101819 seconds
```

Task 3: Trapezoidal Rule

```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/A4$ ./numIntegrate
for t in 1 2 5 7 9; do ./trapT $t | grep Time; done
for t in 1 2 5 7 9; do ./omp_trap $t | grep Time; done
Sequential trapezoidal rule, n = 1000000000
Estimated pi (integral from 0.0 to 1.0) = 3.141592653590 (actual 3.141592653590)
Time taken: 0.774152 seconds
Time taken: 0.752741 seconds
Time taken: 0.391271 seconds
Time taken: 0.170740 seconds
Time taken: 0.136080 seconds
Time taken: 0.123060 seconds
Time taken: 0.777299 seconds
Time taken: 0.389603 seconds
Time taken: 0.189207 seconds
Time taken: 0.137921 seconds
Time taken: 0.122611 seconds
```

Task 4: N Queens

```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/A4$ ./mainQ
for t in 1 2 5 7 9; do ./nqueensT $t | grep Time; done
for t in 1 2 5 7 9; do ./omp_nqueens $t | grep Time; done
Sequential N Queens, N = 15
Solutions = 2279184 (correct)
Time taken: 13.542302 seconds
Time taken: 13.331887 seconds
Time taken: 6.964505 seconds
Time taken: 2.889176 seconds
Time taken: 2.282217 seconds
Time taken: 1.861843 seconds
Time taken: 13.496250 seconds
Time taken: 6.848313 seconds
Time taken: 3.142438 seconds
Time taken: 2.489881 seconds
Time taken: 2.120354 seconds
```

Task 5: Benchmarks

All time in seconds. Sequential and pthread timed with clock_gettime(CLOCK_MONOTIC), OpenMP with omp_get_wtime(); only the computation is timed

Array Sum (N = 1,000,000,000)

	Sequential	1 Thread	2 Threads	5 Threads	7 Threads	9 Threads
Time	.234					
pthread		.227	.116	.081	.080	.080
OpenMP		.219	.111	.080	.078	.080

Matrix Multiplication (matrix size 1,000)

	Sequential	1 Thread	2 Threads	5 Threads	7 Threads	9 Threads
Time	.567					
pthread		.481	.259	.132	.119	.108
OpenMP		.484	.247	.123	.108	.102

Trapezoidal Rule Integration (interval size 1,000,000,000)

	Sequential	1 Thread	2 Threads	5 Threads	7 Threads	9 Threads
Time	.774					
pthread		.753	.391	.171	.136	.123
OpenMP		.777	.390	.189	.138	.123

N Queens (N = 15)

	Sequential	1 Thread	2 Threads	5 Threads	7 Threads	9 Threads
Time	13.542					
pthread		13.332	6.965	2.889	2.282	1.862
OpenMP		13.496	6.848	3.142	2.490	2.120

Task 6: Scheduling

```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/A4$ for s in static dynamic,1 dynamic,1000 guided; do echo $s; OMP_SCHEDUL
E=$s ./omp_nqueens 4 | grep Time; done
static
Time taken: 3.727970 seconds
dynamic,1
Time taken: 3.465838 seconds
dynamic,1000
Time taken: 13.637019 seconds
guided
Time taken: 3.482181 seconds
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/A4$ for s in static dynamic,1 dynamic,1000 guided; do echo $s; OMP_SCHEDUL
E=$s ./omp_sum 4 | grep Time; done
static
Time taken: 0.131715 seconds
dynamic,1
Time taken: 8.015586 seconds
dynamic,1000
Time taken: 0.148858 seconds
guided
Time taken: 0.081850 seconds
```